

Algorithm & Evidence Map for the Feature Roadmap

Every unbuilt feature on the Vytal roadmap, matched to a verified research paper or standard, with the implementation logic written out line by line and mapped back to the codebase.

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How to read this dossier

This document is the companion to the Vytal Feature Roadmap. For every feature that is not yet built, three questions get answered: what it actually does, which specific published paper or standard it rests on, and exactly how to implement it against Vytal's real codebase (`uncertainty.js`, `ai.js`, `ScanPage.jsx`, `ReportPage.jsx`, the `beatTimesMs` array).

Every citation in this dossier was independently re-searched, not carried over unchecked from the original roadmap draft. Where a paper checked out, its DOI or a direct open-access link is given. Where a roadmap citation could not be located under its stated title, that is said plainly, and a verified alternative source is substituted. A small number of features carry no papers at all — these are pure UX or engineering work, and are labelled as such rather than padded with citations that don't apply.

The full algorithm write-up, with pseudocode and honest limitations for each feature, lives in the accompanying `mds/` folder — one Markdown file per feature. This PDF is the map; the Markdown files are the territory.

A note on what "not medical advice" means here. Every algorithm in this dossier is a screening proxy, not a diagnosis. The literature is cited honestly, including where it falls short of clinical-grade accuracy — this dossier treats that as information the team needs, not a marketing problem to write around.

1--7. Built on signal Vytal already has

Every feature below reuses data Vytal's rPPG pipeline already computes for heart rate — no new capture mode, no new hardware.

01. SpO2 Proxy Estimation

Evidence: Ni et al., *ReViSe* (arXiv:2206.08748) — SpO2 MAE 1.64% from facial rPPG. IEEE EMBC 2023 DNN fingertip study — MAE 1.97%. Samsung Research normalisation study for cross-user drift.

Logic: Ratio-of-Ratios on the red vs. blue/green channel amplitude already extracted before CHROM/POS runs. Gated by `uncertainty.js`'s existing SNR check.

Limitation: Needs local calibration against a real oximeter; least tested below 90% SpO2 — the range that matters clinically.
Full detail: [mds/01-spo2-proxy.md](#)

02. Irregular Heartbeat Flag (AFib Proxy)

Evidence: FibriCheck validation (JMIR mHealth) — sensitivity/specificity both >90% vs. 12-lead ECG. UCL systematic review & meta-analysis confirms generalisation across apps.

Logic: RMSSD + pnn50 irregularity check directly on the `beatTimesMs` array Vytal's peak detector already produces — pure downstream analysis, no new signal.

Limitation: Ectopic beats mimic early AFib on a short window; require an 8-beat pattern, not one outlier pair.
Full detail: [mds/02-afib-flag.md](#)

03. Tachycardia / Bradycardia 3-Level Alert Scale

Evidence: WHO IMCI danger-sign thresholds, AHA/ACC tachycardia guidelines, Singer et al. Sepsis-3 (*JAMA* 315(8):801-810, 2016).

Logic: Combined HR+BR+stress threshold table replacing the binary flag, with a Sepsis-3-style severity bump when HR and BR are elevated together.

Limitation: Population thresholds, not personalised — needs pediatric/pregnancy context to be meaningful.
Full detail: [mds/03-alert-scale.md](#)

04--05. Scan History Sparkline & Scan Quality Score

Evidence: None needed — pure UX, surfacing data `uncertainty.js` and the patient record store already compute.

Logic: Inline SVG trend line on the dashboard; one confidence field added to the printed report.

Full detail: [mds/04-scan-history-sparkline.md](#), [mds/05-scan-quality-score.md](#)

06--07. Voice Readout & Auto Re-scan

Evidence: W3C Web Speech API (standard, zero dependency). WHO health-literacy guidance recommends audio delivery for low-literacy populations.

Logic: `window.speechSynthesis` reads the existing multilingual explanation aloud; auto-retry with a fingertip-mode suggestion when `uncertainty.js` flags `reliable: false`.

Full detail: [mds/06-voice-readout.md](#), [mds/07-auto-rescan.md](#)

8--11. New scan modes, real clinical literature

08. Anemia Screening (Conjunctival Pallor)

Evidence (verified): Zhao, L. et al., *Prediction of anemia in real-time using a smartphone camera processing conjunctival images*, *PLOS ONE* 19(5):e0302883 (2024), DOI: 10.1371/journal.pone.0302883. AUC **0.92** at 7g/dL, **0.90** at 9g/dL transfusion thresholds; 435 ED patients.

Logic: RAW-format conjunctival capture → high hue ratio (HSV) → regression to Hb → WHO-band classification (not a raw number).

Not verified: the roadmap's "AnemiaVision" and "3D-printed spacer" citations could not be confirmed.
mds/08-anemia-screening.md

09. Jaundice Screening (Scleral Icterus)

Evidence (verified): Mariakakis, A. et al., *BiliScreen*, *ACM IMWUT* 1(2), Art. 20 (2017), DOI: 10.1145/3090085 — sensitivity 89.7%, specificity 96.8% (70-person study, with light-control accessory). *PLOS ONE* ambient-subtracted sclera chromaticity paper — software-only alternative to a physical light box.

Logic: Two-shot flash/no-flash capture (front camera) → ambient subtraction → MediaPipe eye landmarks (already wired in Vytal) → sclera hue shift → binary "case of concern" flag.

mds/09-jaundice-screening.md

10. Pediatric Mode

Evidence: WHO IMCI age-banded HR/RR reference tables; AHA PALS guidelines.

Logic: Age-group selector swaps the threshold table feeding feature 03's alert scale, and is passed into the Qwen prompt for age-appropriate explanation register.

mds/10-pediatric-mode.md

11. Blood Pressure Estimation (Pulse Transit Time)

Evidence: Ni et al., *ReViSe* — BP MAE 6–10mmHg from facial rPPG. NIH/PubMed systematic review of cuffless BP: per-user calibration required for accuracy. *Frontiers in Physiology* survey notes lack of standardised validation protocols.

Logic: Face scan + fingertip scan → time delay between matched beats in both `beatTimesMs` traces → linear PTT-to-BP mapping, **calibrated** against one real cuff reading per patient.

Limitation: 6–10mmHg MAE fails the IEEE/AAMI <5mmHg clinical standard — trend indicator only, never a standalone reading.
mds/11-blood-pressure-ptt.md

12--14. Combination logic and a new capture mode

12. Respiratory Distress Indicator

Evidence: WHO IMCI pneumonia danger sign (age-banded RR thresholds). Singer et al., Sepsis-3 (*JAMA* 2016) — RR>22 + HR>90 as combined early-sepsis criteria.

Logic: if hr>100 and br>25: flag "distress" — pure combination logic on HR/BR Vytal already outputs, sharing the age-band table introduced for pediatric mode.

mds/12-respiratory-distress.md

13. Pregnancy Mode

Evidence: Established obstetric cardiovascular-adaptation physiology — elevated cardiac output, reduced vascular resistance in the third trimester shift resting HR to roughly 80-100bpm, RR to roughly 16-20/min.

Logic: Toggle swaps the threshold table (same pattern as pediatric mode) feeding the alert scale.

Not verified: no specific published RMSSD-in-pregnancy constant found — stress-baseline shift needs local validation, not a hardcoded number.

mds/13-pregnancy-mode.md

14. Malnutrition Screening Proxy (BMI from Photo)

Evidence (verified): Noy, T.T. et al., *SinBMI* (Technion) — EfficientNet-B2, mobile-optimised. *Digital Scale* (arXiv:2508.20534) — MAPE 7.9% on 71k+ curated images, open-source. *PatchBMI-Net* (arXiv:2311.18102) — facial-patch alternative needing no full-body photo, 5.4× smaller than ResNet-50.

Logic: Standardised full-body capture → lightweight CNN backbone → WHO malnutrition-risk band, paired with existing HR trend data.

Not verified: the roadmap's "ResNet-50" and "SAM Photo Diagnosis App" citations could not be confirmed.

mds/14-malnutrition-bmi.md

15--20. Standards, not studies

These features need protocol compliance, not literature review. Grouped in one dossier entry since none require verifying a research claim.

FHIR Export

HL7 FHIR R4 `Observation` resource, LOINC-coded per vital sign — what lets a Vytal reading import into a hospital EMR.

DHIS2 Integration

DHIS2 Web API v2 Tracker Events — the standard way a point-of-care tool feeds a national health information system.

SMS Fallback & WhatsApp Share

RFC 5724 `sms`: URI scheme; WhatsApp's documented `wa.me` Click-to-Chat link — both client-side triggers, no backend needed.

Batch Scan

Pure UX/session-queue feature — no external standard involved.

Version-pin FHIR (R4 vs R5) and the DHIS2 API version at integration time — both have made breaking changes across releases. Full detail: [mds/15-platform-deployment.md](#)

21--26. From single scan to longitudinal signal

21. Longitudinal Patient Risk Scoring

Evidence (verified): *Critical Care Medicine* 49:e961–e967 (2021), DOI: 10.1097/CCM.0000000000005064 — trend-based EWS effectively predicted inpatient deterioration. JMIR 23(2):e25187 (2021) scoping review: aggregate-weighted scores “do not incorporate trends.” Churpek et al., *Resuscitation* 102:1–5 (2016) — rate-of-change predicts deterioration.

Logic: Linear-regression slope over the last 3–5 scans per patient; escalates the alert tier when trend and current reading agree.

Limitation: all cited validation is hourly-interval hospital data — Vytal’s visit-to-visit interval is far sparser; thresholds need local re-derivation.
mds/21-longitudinal-risk-scoring.md

22. Population-Level Anomaly Detection

Not verified: the roadmap’s *Lancet Digital Health* 2021 citation could not be located under its stated title. **Real alternatives found:** sub-Saharan Africa mHealth surveillance systematic review (PMC4245630); a genuine *Lancet Digital Health* 2021 mHealth-data commentary (PIIS2589-7500(21)00212-0).

Logic: Server-side, privacy-preserving aggregation of flag rates per region vs. that region’s own 90-day rolling baseline; statistical process control trigger.

Limitation: the least concretely research-backed feature in this roadmap — treat as emerging, not validated.
mds/22-population-anomaly-detection.md

23--26. Clinician View, Coaching, Programme Context, Language Expansion

Evidence: None needed — all four extend Vytal’s already-built Qwen prompt layer via new prompt templates and configuration, not new algorithms.

mds/23-ai-layer-clinician-coaching-language.md

27--29. Optional accessories, real device targets

27. Bluetooth Pulse Oximeter Pairing

Evidence: W3C Web Bluetooth API; Bluetooth SIG Pulse Oximeter Service (PLXS) GATT profile — the standard characteristic layout exposed by compliant devices including Wellue O2Ring, Nonin WristOx.

Logic: `navigator.bluetooth.requestDevice` → GATT connect → parse PLXS Spot-Check/Continuous Measurement characteristics. A paired reading captured alongside a camera scan is exactly the reference data feature 01's SpO2 calibration needs.

Limitation: Web Bluetooth is unsupported on Safari/iOS.

mds/27-ble-pulse-oximeter.md

28. Thermal Camera (USB-C)

Evidence (verified): *Diagnostic accuracy of non-contact infrared thermometers and thermal scanners: a systematic review and meta-analysis, Journal of Travel Medicine* 27(8):taaa193 (2020) — forehead NCIT sensitivity **0.808**, specificity **0.920**; thermal scanners sensitivity **0.818**, specificity **0.923**.

Logic: FLIR Lepton 3.5 (\$79, USB-C, 160×120 @ 8.6Hz) via Web USB → frame parsed per FLIR's IDD → forehead ROI located via Vytal's existing MediaPipe landmarks → fever threshold combined with the respiratory-distress flag (feature 12).

Correction: the roadmap's "84%/91%" figure is close to but not exactly the verified meta-analysis numbers above — use the precise range.

mds/28-thermal-camera.md

29. Wearable HRV Baseline

Not verified: the roadmap's *Frontiers in Psychology* 2022 citation could not be located. **Real alternatives found:** Shaffer & Ginsberg, *Frontiers in Public Health* 5:258 (2017) — HRV norms depend heavily on measurement context. *Frontiers in Neuroscience* 14:564123 (2020) — personalised per-participant calibration outperformed fixed population cutoffs.

Logic: Garmin/HealthKit/Fitbit SDK pulls trailing 14-30-day RMSSD as a personal baseline; scan-time RMSSD compared against that baseline (z-score) instead of a population cutoff.

mds/29-wearable-hrv-baseline.md

SUMMARY

Evidence strength, at a glance

Feature	Strength	Key check
SpO2 Proxy	Strong	IEEE EMBC 2023, ReViSe arXiv
AFib Flag	Strong	JMIR FibriCheck, UCL meta-analysis
Anemia Screening	Strong	PLOS ONE 2024, AUC 0.90-0.92 (verified DOI)
Jaundice Screening	Strong	BiliScreen ACM IMWUT 2017 (verified DOI)
Blood Pressure (PTT)	Moderate	ReViSe arXiv; MAE exceeds clinical standard
BMI / Malnutrition	Moderate	SinBMI, Digital Scale arXiv (verified)
Respiratory Distress	Protocol	WHO IMCI, Sepsis-3 JAMA 2016
Pediatric Mode	Protocol	WHO IMCI, AHA PALS
Thermal Fever Screen	Strong	J. Travel Med. 2020 meta-analysis (verified)
Longitudinal Risk	Emerging	Crit Care Med 2021 (verified), sparse-sampling gap
Population Anomaly	Unverified	original citation not locatable; real alternatives substituted
Wearable HRV Baseline	Unverified	original citation not locatable; real alternatives substituted

What this means for the pitch. Anemia and jaundice screening are the strongest unbuilt features — both have a real, DOI-verified paper with a specific, checkable accuracy number behind them. Population anomaly detection and the wearable HRV baseline are the two features whose original citations didn't hold up under checking; that doesn't mean the ideas are wrong, it means they should be pitched as directions worth prototyping, not as already-validated the way the anemia and jaundice screens can be.

SOURCES.

Every DOI, arXiv ID and open-access link cited in this dossier, in one place.

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